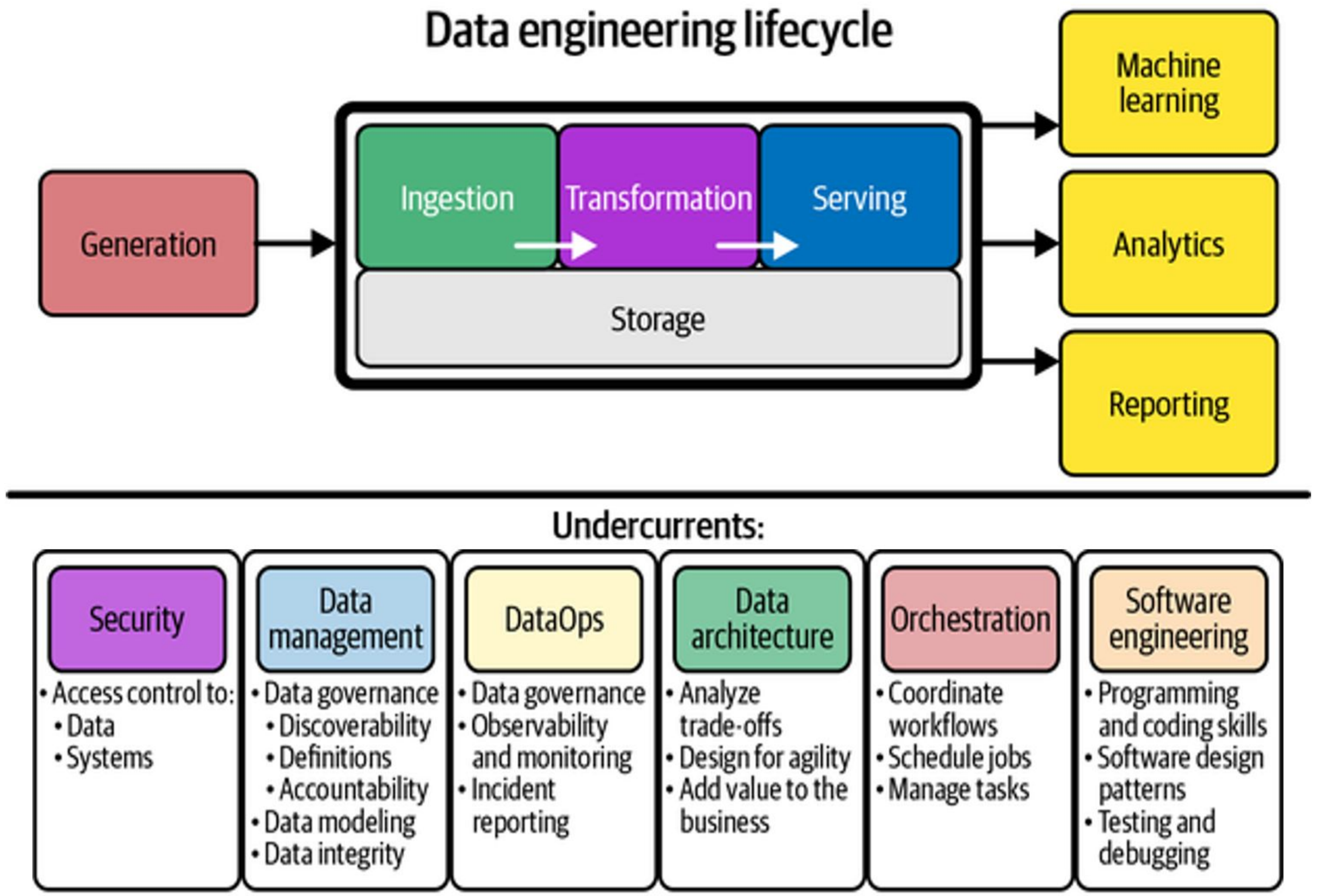
Assignment 2

Context

Assignments in Big Data Technologies pursue a holistic understanding and management of the data lifecycle in general and the data engineering lifecycle in particular.

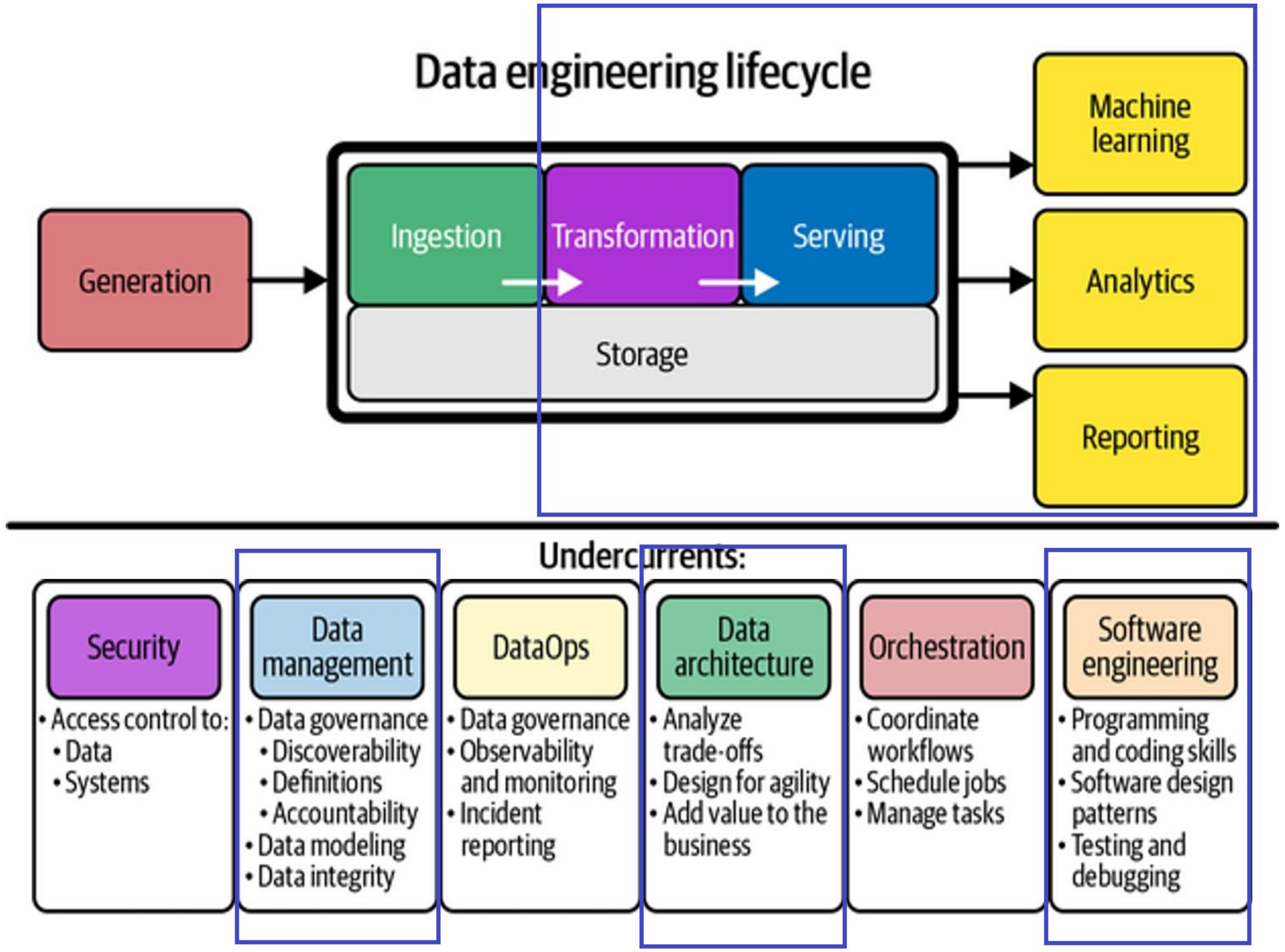
The student will face a situation where generating value from data is not obvious. Leveraging big data technologies will be the way to manage the questions that will come up before serving intelligence with trust.



Transforming and Serving Data

The Assignment 2 focuses on the right side of the data lifecycle: transforming and serving data. When we combine this two tasks the result is real value in the form of a Machine learning model, a Dashboard or a report from Analytics.

This assignment aims to validate that the student can design and implement a reliable data transformation pipeline and generate value.



Statement

Transformation, data management and software engineering

The step of data transformation requires the student to identify the potential value in the dataset generating questions around it in the first place. The student will integrate the dataset in the Data Warehouse to transform the data in one place to answer the proposed questions. The requirements for this task are:

- Use dbt to control data transformations.
- Generate a lineage diagram with all the software components (models) designed to achieve the end-to-end project.
- Generate a data dictionary explaining the metrics and the transformations generated to answer the business questions.

Data architecture design

The student will design a diagram of the architecture that will enable the data integration end-to-end, from the OLTP systems to the Datawarehouse. The systems, tools or services are of free choice. Open source is recommended but not mandatory.

In addition to the Architecture diagram, the student will explain the reason behind the selection of each component and the benefits of using them regarding reliability, scalability and maintainability of the entire system and the resources available alongside the identification of possible roles associated with each involved process.

Statement

Brief of the assignment

- Business or research questions that we are aiming to answer with data
- The resulting answers of such analysis in the form of a dashboard, report or model
- The data lineage generated to get to business answers
- Conclusions and further work

Deliverables

- 10-page long report explaining each required issue containing transformations and resulting data assets like data models, dashboards or ml models generated.
- The final presentation that will be used in the presentation in class in .pdf format.
- Source code of each batch process, query, script, or programmed used throughout the process

Delivery date

- 27th of April 2025, 17:00.